STPM 2019 PHYSICS 960/1

Questions and Answers

All calculations use $g = 9.81\ m\ s^{-2}$ and the values of constants printed in the question paper.
Question 2 has no correct option; it was not counted in the examination.

Section A [15 marks]

1. The dimensions of $\frac{\text{pressure}}{\text{density}}$ are

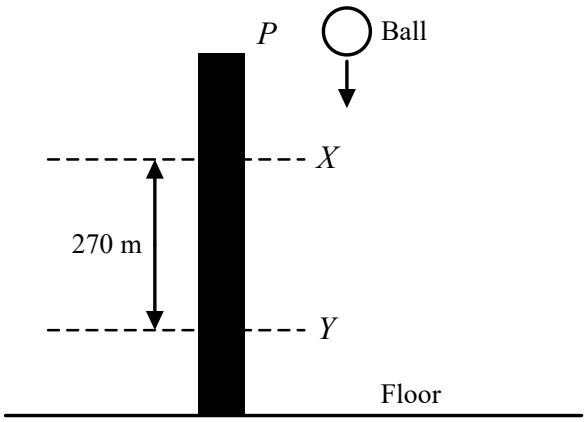
A LT^{-1} B L^2T^{-2} C ML^{-3} D $ML^{-1}T^{-2}$

Answer: B

$$\left[\frac{\text{pressure}}{\text{density}}\right] = \frac{ML^{-1}T^{-2}}{ML^{-3}} = L^2T^{-2}$$

Explanation: Pressure = force/area has dimensions $MLT^{-2}/L^2 = ML^{-1}T^{-2}$, and density = mass/volume is ML^{-3}. The result has the dimensions of (speed)^2, consistent with $c^2 = 3p/\rho$.

2. A ball is released from P as shown in the diagram below.



The time taken for the ball to travel from P to X is 3.0 s. If the distance between X and Y is 270 m, what is the time taken by the ball to travel from X to Y?

A 6.0 s B 6.8 s C 7.4 s D 8.0 s

Answer: none of the options (5.0 s)

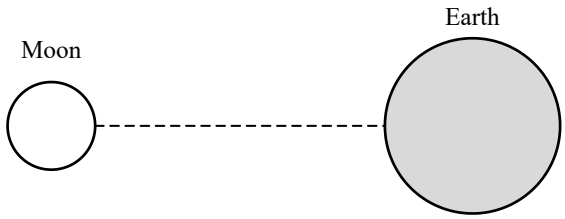
$$\text{At } X\text{: } v = gt = (9.81)(3.0) = 29.4\ \text{m s}^{-1}$$

$$270 = 29.4t + \frac{1}{2}(9.81)t^2 \Rightarrow 4.905t^2 + 29.4t - 270 = 0$$

$$t = 5.0\ \text{s}$$

Explanation: The correct working gives 5.0 s, which is not one of the options, so this question was not counted. The method is still useful: find the speed at X after 3.0 s of free fall, then solve the quadratic for the time to fall the next 270 m.

3. The Earth-moon system is shown in the diagram below.



Which statement is true about the centre of mass of the Earth-moon system?

A The centre of mass is nearer to the Earth.
B The centre of mass rotates around the moon.
C The centre of mass is acted on by an external force.
D The centre of mass does not lie on the line joining the Earth to the moon.

Answer: A

The Earth is much more massive than the moon, so the centre of mass of the system lies on the line joining them, much nearer to the Earth (in fact inside the Earth).

Explanation: For two bodies, $m_E r_E = m_M r_M$, so the distance of the centre of mass from each body is inversely proportional to its mass. Both the Earth and the moon orbit the centre of mass.

4. A stone with a mass of 345 kg is placed on an adjustable plane. A worker has raised one end of the plane to move the stone down the plane. The angle of inclination of the plane is 30° when the stone starts to move. What is the coefficient of static friction between the stone and the surface of the inclined plane?

A 0.58 B 0.87 C 1.0 D 1.7

Answer: A

On the point of sliding: $mg \sin 30° = \mu_s mg \cos 30°$

$\mu_s = \tan 30° = 0.58$

Explanation: The mass of the stone (345 kg) is not needed: at the angle where the stone just starts to slide, the coefficient of static friction equals the tangent of the angle.

5. A block with a mass of 0.20 kg is hung at the end of a spring and it is lifted vertically so that the spring is unstretched. The block is then released and it touches the table with a velocity of 0.80 m s^{-1}. If the distance of the table from the released point of the spring is 0.10 m, what is the force constant of the spring?

A 13 N kg^{-1} B 26 N kg^{-1} C 39 N kg^{-1} D 52 N kg^{-1}

Answer: B

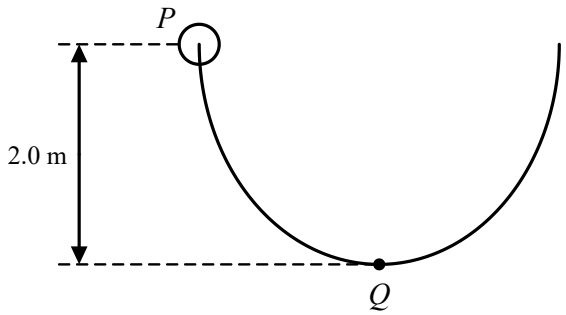
$$\text{Energy: } mgh = \frac{1}{2}kx^2 + \frac{1}{2}mv^2, \text{ with } x = h = 0.10 \text{ m}$$

$$(0.20)(9.81)(0.10) = \frac{1}{2}k(0.10)^2 + \frac{1}{2}(0.20)(0.80)^2$$

$$0.196 = 0.005k + 0.064 \Longrightarrow k = 26 \text{ N m}^{-1}$$

Explanation: The gravitational potential energy lost becomes elastic potential energy in the spring plus the kinetic energy of the block. The extension of the spring equals the distance fallen. (The unit in the options should read N m^{-1}.)

6. A marble is released from *P* on a smooth surface of a fixed semi-circular pipe with a radius of 2.0 m as shown in the diagram below.



If the mass of the marble is 500 g, what is the centripetal force on the marble at *Q*?

A 0 B 4.9 N C 9.8 N D 39 N

Answer: C

$$\frac{1}{2}mv^2 = mgh \Longrightarrow v^2 = 2(9.81)(2.0) = 39.2 \text{ m}^2 \text{ s}^{-2}$$

$$F = \frac{mv^2}{r} = \frac{(0.500)(39.2)}{2.0} = 9.8 \text{ N}$$

Explanation: The pipe is smooth, so the speed at *Q* comes from conservation of energy. The centripetal force is $mv^2/r = 2mgh/r$; with $h = r$ this is $2mg$. Remember 500 g = 0.500 kg.

7. The gravitational field strength on the surface of the moon is 1.6 m s^{-2}. If the moon is assumed to be a sphere with a radius of 1.7×10^6 m, what is the mass of the moon?

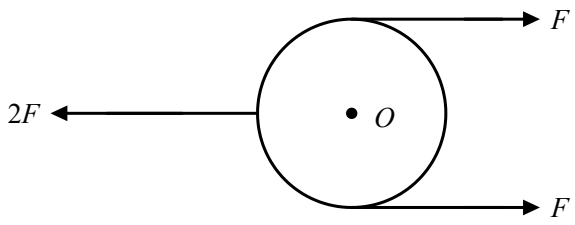
A 2.0×10^8 kg B 2.9×10^8 kg C 4.1×10^{16} kg D 6.9×10^{22} kg

Answer: D

$$g = \frac{GM}{R^2} \Longrightarrow M = \frac{gR^2}{G} = \frac{(1.6)(1.7 \times 10^6)^2}{6.67 \times 10^{-11}} = 6.9 \times 10^{22} \text{ kg}$$

Explanation: The gravitational field strength at the surface is GM/R^2. Remember to square the radius.

8. Three forces, each with a magnitude of F, F and $2F$, act on a spherical object of centre O are shown in the diagram below.



Which statement is true about the spherical object?

A The spherical object moves with a linear acceleration.
B The spherical object moves to the left and then stops.
C The spherical object is stationary.
D The spherical object rotates.

Answer: C

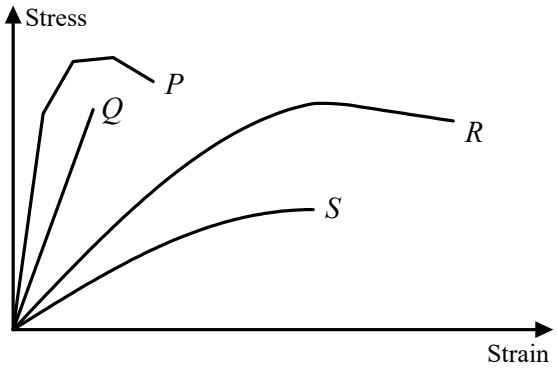
Resultant force: $F + F - 2F = 0$

Torques about O: the upper F (clockwise) and the lower F (anticlockwise) have equal moment arms, so they cancel; $2F$ acts through O and has no moment.

Both conditions for equilibrium are satisfied, so the object stays stationary.

Explanation: For a rigid body to be in equilibrium, both the resultant force and the resultant torque must be zero. Check each one separately.

9. The stress-strain curves for four different materials, P, Q, R and S, of the same dimensions are shown in the graph below.



Which statement is **not** true about the curves?

A P is the strongest.
B Q is the most brittle.
C R is the most ductile.
D S is the stiffest.

Answer: D

Stiffness is shown by the gradient of the linear part. S has the smallest gradient, so it is the **least** stiff. Statement D is not true.

Explanation: P reaches the highest stress (strongest), Q breaks at the end of the straight line with no plastic region (most brittle), and R has the largest plastic region (most ductile). The steepest initial gradients belong to P and Q.

10. What is the r.m.s. speed of hydrogen molecules at a pressure of 1.5×10^5 Pa and density of 0.14 kg m^{-3}?

A 6.0×10^2 m s^{-1} B 1.8×10^3 m s^{-1} C 3.6×10^5 m s^{-1} D 3.2×10^6 m s^{-1}

Answer: B

$$p = \frac{1}{3}\rho\langle c^2\rangle \Longrightarrow c_{\text{r.m.s.}} = \sqrt{\left(\frac{3p}{\rho}\right)} = \sqrt{\left(\frac{3(1.5 \times 10^5)}{0.14}\right)} = 1.8 \times 10^3 \text{ m s}^{-1}$$

Explanation: Rearrange the kinetic theory pressure equation. No temperature or molar mass is needed when the density is given.

11. At 50 K, a hydrogen molecule has three degrees of freedom. When the temperature is raised to 300 K, the number of degree of freedom increases to five. The increase in the number of degrees of freedom of the molecule is due to

A rotational motion
B vibrational motion
C translational motion
D rotational and vibrational motion

Answer: A

Between 50 K and 300 K the hydrogen molecule gains 2 rotational degrees of freedom (3 translational + 2 rotational = 5).

Explanation: Rotational modes need a minimum amount of energy to become active; at very low temperatures only translational motion is possible. Vibrational modes only become active at much higher temperatures (thousands of kelvin).

12. An ideal gas with an initial volume, V_0, and absolute temperature, T_0, is compressed adiabatically until its volume becomes $\frac{3}{8}V_0$ and the absolute temperature is $\frac{3}{2}T_0$. What is the ratio of the molar heat capacity of the gas?

A 1.33 B 1.41 C 1.67 D 2.50

Answer: B

$$\text{Adiabatic: } TV^{\gamma-1} = \text{constant} \Longrightarrow \frac{3}{2} = \left(\frac{8}{3}\right)^{\gamma-1}$$

$$\gamma - 1 = \frac{\ln 1.5}{\ln 2.667} = 0.413 \Longrightarrow \gamma = 1.41$$

Explanation: The ratio of molar heat capacities is $\gamma = C_{p,m}/C_{V,m}$. Use logarithms to solve for the power.

13. Which statement is true about natural convection and forced convection of fluid?

A The heat transfer is generated by an external heat source.
B The heat transfer is used to increase the rate of heat exchange.
C The heat transfer is by means of the movement of the fluid itself.
D The heat transfer involves the increase of the denser part of the fluid.

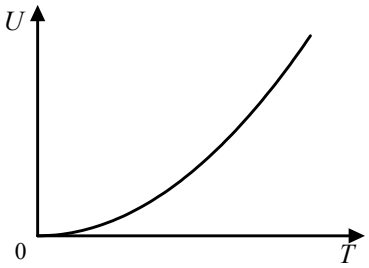
Answer: C

In both natural and forced convection, heat is transferred by the movement of the fluid itself.

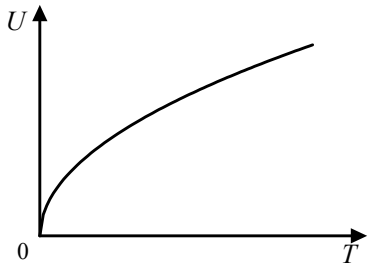
Explanation: In natural convection the fluid moves because of density differences (hot fluid rises); in forced convection a pump or fan moves it. Either way the fluid carries the heat with it.

14. Which graph represents the variation of internal energy, U, with temperature, T, of an ideal gas?

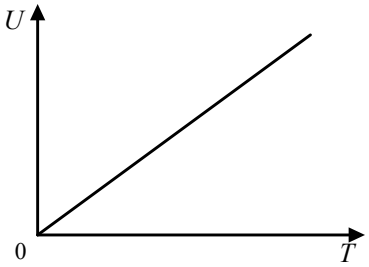
A



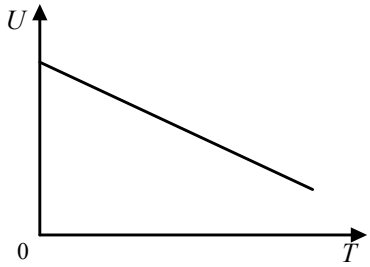
B



C



D



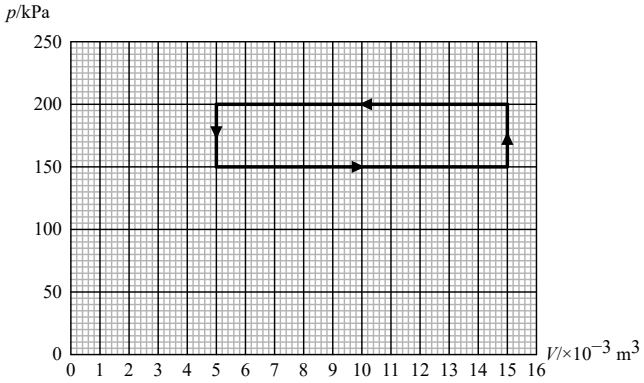
Answer: C

$$U = \frac{f}{2}nRT \Longrightarrow U \propto T$$

A straight line through the origin.

Explanation: The internal energy of an ideal gas is entirely the kinetic energy of its molecules, which is directly proportional to the absolute temperature.

15. A cyclic thermodynamic process which is represented by the *p*–*V* graph is shown below.



What is the net work done on the gas in a cycle?

A 500 J B 1500 J C 2000 J D 3500 J

Answer: A

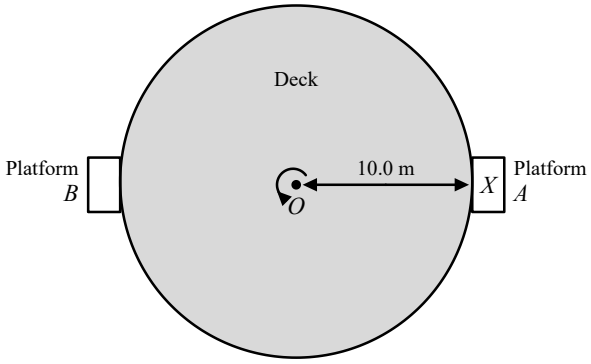
$$\text{Area enclosed} = (200 - 150) \times 10^3 \times (15 - 5) \times 10^{-3} = 500 \text{ J}$$

The cycle is anticlockwise, so net work is done **on** the gas: +500 J.

Explanation: The net work in a cycle is the area of the loop. The gas is compressed at the higher pressure (200 kPa) and expands at the lower pressure (150 kPa), so more work is done on the gas than by it.

Section B [15 marks]

16. A circular deck with a radius of 10.0 m is rotating above the water's surface to make one complete cycle in one minute as shown in the diagram below.



A man, *X*, with a mass of 70.0 kg at platform *A* jumps on the deck to reach platform *B*.

(a) Calculate the angular velocity of the deck. [2]

Answer

$\omega = \frac{2\pi}{T} = \frac{2\pi}{60 \text{ s}}$ 1

$\omega = 0.105 \text{ rad s}^{-1}$ 1

2

Explanation: One complete cycle is 2π radians, and one minute is 60 s.

(b) Calculate the centripetal force experienced by X. [2]

Answer

$F = mr\omega^2 = (70.0)(10.0)(0.1047)^2$ 1

$F = 7.68$ N 1

[7.72 N if ω is rounded to 0.105 rad s^{-1} first]

2

Explanation: X is at the edge of the deck, 10.0 m from the centre. Avoid rounding ω too early, because it is squared.

(c) Calculate the tangential velocity for X at the edge of the deck. [2]

Answer

$v = r\omega = (10.0)(0.1047)$ 1

$v = 1.05$ m s^{-1} 1

2

Explanation: Every point on the deck has the same angular velocity, but the tangential speed is proportional to the distance from the centre.

(d) State the force enable X to remain stationary on the deck. [1]

Answer

Static friction (between the feet of X and the deck), acting towards the centre 1

1

Total for question **7**

Explanation: Static friction supplies the centripetal force that keeps X moving in a circle with the deck, so he does not slide relative to it.

17. A double-paned window has a thin layer of airtight seal between two panes of glass which is often used to minimise the transmission of heat through the glass panes.

(a) Explain the mechanism of heat conduction through the double-paned window. [2]

Answer

Glass and air are non-metals: atoms at the hotter side vibrate more vigorously (with larger amplitude) — 1

and collide with neighbouring atoms/molecules, passing energy along to the cooler side; air is a poor conductor because its molecules are far apart — 1

2

Explanation: There are no free electrons in glass or air, so conduction happens only by vibrations and collisions. The air gap, being a very poor conductor, causes most of the reduction in heat flow.

(b) Why the layer of air must be thin? [1]

Answer

So that convection currents cannot be set up in the air layer (heat is transferred only by conduction) — 1

1

Explanation: In a wide gap the air could circulate, carrying heat across by convection. A thin layer keeps the air still.

(c) A double-paned window with an area of 6.0 m^2 is consisting of two panes of glass, each with a thickness of 4.00 mm, separated by a layer of air with a thickness of 5.00 mm. If the temperature inside the house is 20 °C and temperature outside is 35 °C,

(i) calculate the rate of heat flow into the house, [3]

Answer

In series, same rate of heat flow: $\frac{dQ}{dt} = \frac{A(\theta_1 - \theta_2)}{\Sigma(x/k)}$ — 1

$\frac{dQ}{dt} = \frac{(6.0)(35-20)}{2(\frac{0.00400}{0.800}) + \frac{0.00500}{0.025}} = \frac{90}{0.010 + 0.200}$ — 1

Rate of heat flow = 429 W — 1

3

Explanation: Add the thermal resistances x/k of the two glass panes and the air layer. The air layer (0.200) is twenty times the resistance of the glass (0.010), so it controls the heat flow.

(ii) determine the percentage of reduction in the rate of heat flow by the double-paned window compared to a single-paned window.

[Thermal conductivity of glass is 0.800 W m^{-1} K^{-1}, thermal conductivity of air is 0.025 W m^{-1} K^{-1}.] [2]

Answer

Single pane: $\frac{dQ}{dt} = \frac{(0.800)(6.0)(15)}{0.00400} = 1.80 \times 10^4$ W — 1

Percentage reduction = $\frac{18\,000 - 429}{18\,000} \times 100\% = 97.6\%$ — 1

2

Total for question **8**

Explanation: Compare with one 4.00 mm pane of glass under the same temperature difference. The percentage reduction is the decrease divided by the single-pane value.

Section C [30 marks]

18. (a) Define Newton's law of universal gravitation. [1]

Answer

Every particle in the universe attracts every other particle with a force that is directly proportional to the product of their masses and inversely proportional to the square of the distance between them. — 1

1

Explanation: In symbols, $F = \frac{Gm_1m_2}{r^2}$.

(b) A satellite is orbiting the Earth in a circular orbit. Derive the relationship between the period, T, and the radius of the orbit, R. [5]

Answer

Gravitational force on the satellite: $F = \frac{GMm}{R^2}$ — 1

It provides the centripetal force: $\frac{GMm}{R^2} = mR\omega^2$ — 1

$\omega = \frac{2\pi}{T} \Rightarrow \frac{GM}{R^2} = \frac{4\pi^2 R}{T^2}$ — 1

$T^2 = \frac{4\pi^2}{GM} R^3$ — 1

$T^2 \propto R^3$ (Kepler's third law), since $4\pi^2/GM$ is constant — 1

5

Explanation: Finish the derivation with a clear conclusion: $T^2 \propto R^3$, or $T \propto \sqrt{(R^3)}$. The mass of the satellite cancels, so the period depends only on the orbit radius and the mass of the planet.

(c) A satellite with a mass of 1337 kg is initially orbiting the planet Mars at an altitude of 45 000 km, gradually descending to a lower altitude of 316 km above the surface of the planet Mars. Calculate the

(i) potential energy of the satellite at the initial altitude, [3]

Answer

$r_1 = 3.39 \times 10^6 + 45\,000 \times 10^3 = 4.839 \times 10^7$ m — 1

$U = -\frac{GMm}{r_1} = -\frac{(6.67 \times 10^{-11})(6.42 \times 10^{23})(1337)}{4.839 \times 10^7}$ — 1

$U = -1.18 \times 10^9$ J — 1

3

Explanation: The distance in the formula is measured from the centre of Mars, so add the radius of Mars to the altitude (in metres). The potential energy is negative.

(ii) change in energy of the satellite during the descend, [3]

Answer

Total energy in orbit: $E = -\frac{GMm}{2r}$; $r_2 = 3.39 \times 10^6 + 316 \times 10^3 = 3.706 \times 10^6$ m — 1

$\Delta E = -\frac{GMm}{2}(\frac{1}{r_2} - \frac{1}{r_1}) = -7.724 \times 10^9 - (-5.916 \times 10^8)$ — 1

$\Delta E = -7.13 \times 10^9$ J — 1

3

Explanation: The change in energy of the satellite is the change in its **total** energy (kinetic + potential). In a lower orbit the total energy is more negative: the satellite loses energy (for example to air resistance), although its kinetic energy increases.

(iii) orbital speed of the satellite after descending to the altitude of 316 km.
[Mass of the planet Mars is 6.42×10^{23} kg and radius of the planet Mars is 3.39×10^6 m.] [3]

Answer

$\frac{GMm}{r^2} = \frac{mv^2}{r} \Rightarrow v = \sqrt{(\frac{GM}{r_2})}$ — 1

$v = \sqrt{(\frac{(6.67 \times 10^{-11})(6.42 \times 10^{23})}{3.706 \times 10^6})}$ — 1

$v = 3.40 \times 10^3$ m s^{-1} — 1

3

Total for question **15**

Explanation: The gravitational force provides the centripetal force. Use the orbit radius from the centre of Mars.

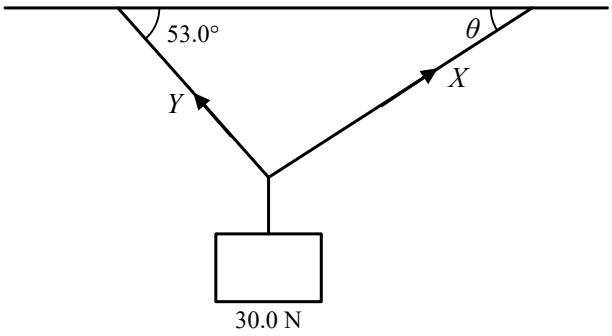
19. (a) State the conditions for equilibrium of a rigid body. [2]

Answer

The resultant (vector sum of) external forces is zero	1
The resultant torque about any point is zero	1
	2

Explanation: The first condition gives translational equilibrium (no linear acceleration); the second gives rotational equilibrium (no angular acceleration).

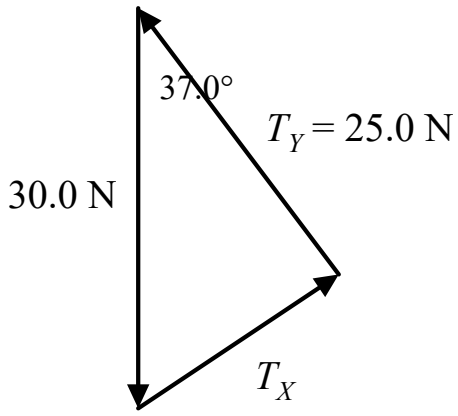
(b) Two light wires, X and Y, hung from a ceiling to support a load of 30.0 N is shown in the diagram below.



Y makes an angle of 53.0° with the ceiling and the tension of 25.0 N. While X makes an angle of θ with the ceiling.

(i) Sketch and label the triangle of forces. [2]

Answer



Closed triangle with the arrows following each other in the same sense (head to tail)	1
All three forces labelled: 30.0 N, T_X and $T_Y = 25.0$ N	1
	2

Explanation: Three forces in equilibrium form a closed triangle when drawn head to tail, each side parallel to its force.

(ii) Determine θ and the tension of X. [4]

Answer

Horizontal: $T_X \cos\theta = 25.0 \cos 53.0° = 15.05$ N — 1

Vertical: $T_X \sin\theta + 25.0 \sin 53.0° = 30.0 \Longrightarrow T_X \sin\theta = 10.03$ N — 1

$\tan\theta = \dfrac{10.03}{15.05} \Longrightarrow \theta = 33.7°$ — 1

$T_X = \sqrt{(15.05^2 + 10.03^2)} = 18.1$ N — 1

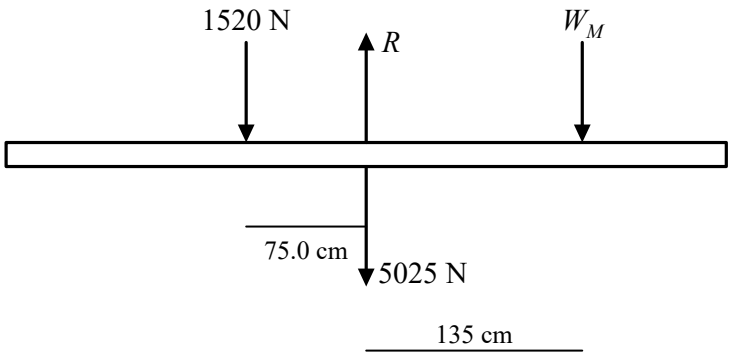
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Explanation: Resolve all the forces horizontally and vertically. The horizontal components of the two tensions balance each other, and the vertical components together support the 30.0 N load.

(c) A uniform plank with a weight of 5025 N and length of 450 cm is pivoted at its centre of mass. A block of 1520 N is placed 150 cm from one of its end. Another block, M, is placed 90.0 cm from the opposite end of the plank so that the system is in equilibrium.

(i) Draw a free body diagram showing the forces acting on the plank. [1]

Answer



Reaction R upward at the pivot, plank weight 5025 N at the centre, 1520 N and W_M downward at their positions — 1

1

Explanation: The pivot is at the centre of mass, so the plank's own weight acts through the pivot. 1520 N is 225 − 150 = 75.0 cm from the pivot and M is 225 − 90.0 = 135 cm from the pivot on the other side.

(ii) Calculate the weight of M. [2]

Answer

Principle of moments about the pivot: $1520 \times 75.0 = W_M \times 135$ — 1

$W_M = 844$ N — 1

2

Explanation: Taking moments about the pivot removes both R and the weight of the plank, which act through the pivot.

(iii) Determine the reaction force acting at the pivot. [2]

Answer

$R = 5025 + 1520 + 844$ — 1

$R = 7.39 \times 10^3$ N — 1

2

Explanation: The plank is in equilibrium, so the upward reaction balances all the downward forces, including the plank's own weight.

(iv) Determine the net torque about the pivot if the 1520 N block is moved 25.0 cm from its original position, away from the pivot. [2]

Answer

New distance of 1520 N = 75.0 + 25.0 = 100 cm; net torque = 1520(1.00) − 844.4(1.35) — 1

Net torque = 380 N m (towards the 1520 N side) — 1

[381 N m if W_M = 844 N is used]

2

Total for question **15**

Explanation: Moving the block further from the pivot increases its moment, so the moments no longer balance and the plank turns towards the 1520 N block.

20. (a)

(i) Define molar heat capacity of gas at constant volume, $C_{V,m}$, and molar heat capacity of gas at constant pressure, $C_{p,m}$. [2]

Answer

$C_{V,m}$: the quantity of heat required to raise the temperature of one mole of the gas by 1 K at constant volume	1
$C_{p,m}$: the quantity of heat required to raise the temperature of one mole of the gas by 1 K at constant pressure	1
	2

Explanation: Both definitions need "one mole", "1 K (or 1 °C)" and the condition (constant volume or constant pressure). Unit: J mol^{-1} K^{-1}.

(ii) Explain why the value of $C_{p,m}$ is bigger than that of $C_{V,m}$. [2]

Answer

At constant volume no work is done, so all the heat supplied increases the internal energy	1
At constant pressure the gas expands and does work against the external pressure, so extra heat is needed for the same 1 K rise	1
	2

Explanation: The increase in internal energy for a 1 K rise is the same in both cases; the difference $C_{p,m} - C_{V,m}$ is the work done in the expansion.

(b) Using the first law of thermodynamics, derive the relationship between molar heat capacity at constant pressure, $C_{p,m}$, for a gas, its molar heat capacity at constant volume, $C_{V,m}$, and molar gas constant, R. [6]

Answer

First law: $Q = \Delta U + W$	1
Constant volume: $W = 0$, so $Q_V = \Delta U = nC_{V,m}\Delta T$	1
For an ideal gas U depends only on T, so $\Delta U = nC_{V,m}\Delta T$ for any process with the same ΔT	1
Constant pressure: $nC_{p,m}\Delta T = \Delta U + p\Delta V$	1
$pV = nRT \Longrightarrow p\Delta V = nR\Delta T$ at constant pressure	1
$nC_{p,m}\Delta T = nC_{V,m}\Delta T + nR\Delta T \Longrightarrow C_{p,m} - C_{V,m} = R$	1
	6

Explanation: The key step is that the change in internal energy is the same for the same temperature rise, whether the heating is at constant volume or at constant pressure.

(c) A cylinder contains 1.00×10^{22} molecules of monatomic gas at the temperature of 10.0 °C. The gas is heated to 80.0 °C. Calculate the amount of heat absorbed by the gas if the process is done at constant pressure. [5]

Answer

$n = \frac{N}{N_A} = \frac{1.00 \times 10^{22}}{6.02 \times 10^{23}} = 0.0166$ mol — 1

Monatomic: $C_{V,m} = \frac{3}{2}R$ — 1

$C_{p,m} = C_{V,m} + R = \frac{5}{2}R$ — 1

$Q = nC_{p,m}\Delta T = (0.0166)(\frac{5}{2})(8.31)(80.0 - 10.0)$ — 1

$Q = 24.2$ J — 1

5

Total for question **15**

Explanation: A monatomic gas has 3 degrees of freedom, so $C_{V,m} = \frac{3}{2}R$, and the result of (b) gives $C_{p,m}$. The temperature change is 70.0 K (a change in °C equals a change in K).